Unit 6: Course Project

1. Project Setup

This section ensures your environment is ready for Terraform deployment (HashiCorp, 2023).

Steps:

1. Open Terminal.
2. Create project folder: `mkdir terraform-project`
3. Navigate: `cd terraform-project`
4. Initialize Git (optional): `git init`

```
(base) a1234@1234s-Laptop Group_1 % terraform state list
(base) a1234@1234s-Laptop Group_1 % cd /Users/a1234/Desktop
mkdir Group_1
cd Group_1
```

Image 1: Project setup

2. Create Required Files

Terraform uses modular files for organization (Microsoft, 2024).

Steps:

1. Create files:
   touch main.tf variables.tf terraform.tfvars network.tf vm.tf outputs.tf
2. Create data file: students.csv

```
(base) a1234@1234s-Laptop Group_1 % touch main.tf variables.tf terraform.tfvars network.tf vm.t
f outputs.tf students.csv
(base) a1234@1234s-Laptop Group_1 % terraform init
terraform fmt
terraform validate
terraform plan
terraform apply
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 4.70.0"...
- Finding hashicorp/random versions matching "~> 3.6"...
- Installing hashicorp/azurerm v4.70.0...
- Installed hashicorp/azurerm v4.70.0 (signed by HashiCorp)
- Installing hashicorp/random v3.8.1...
- Installed hashicorp/random v3.8.1 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
Success! The configuration is valid.
```

3. main.tf

Defines provider and Terraform settings.

Steps:

1. Add Azure provider block.
2. Specify required version.
3. Run `terraform init.`

main.tf - .../Desktop/Group_1

network.tf | main.tf

```
terraform {
  required_providers {
    azurerm = {
      source  = "hashicorp/azurerm"
      version = "~> 4.70.0"
    }
    random = {
      source  = "hashicorp/random"
      version = "~> 3.6"
    }
  }
}

provider "azurerm" {
  features {}
}
```

4. variables.tf
Stores reusable variables.

Steps:
1. Define variables like location and vm_size.
2. Use type and default values.

variables.tf - .../Desktop/Group_1

variables.tf

```
variable "resource_group_name" {
  type = string
}

variable "location" {
  type = string
}

variable "admin_username" {
  type = string
}
```

5. terraform.tfvars (Central US)
Assigns values to variables.

Steps:
1. Add location = "Central US".
2. Add resource names.
3. Save file.

terraform.tfvars

```
resource_group_name = "unit6-central-final-rg"
location            = "Central US"
admin_username      = "Group_1"
```

6. network.tf

Defines networking components (Microsoft, 2024).

Steps:

1. Create resource group.
2. Create virtual network.
3. Create subnet.
4. Create NIC

network.tf - .../Desktop/Group_1

network.tf main.tf

```
resource "azurerm_resource_group" "rg" {
  name     = var.resource_group_name
  location = var.location
}

resource "azurerm_virtual_network" "vnet" {
  name                = "unit6-vnet"
  address_space       = ["10.0.0.0/16"]
  location            = azurerm_resource_group.rg.location
  resource_group_name = azurerm_resource_group.rg.name
}

resource "azurerm_subnet" "subnet" {
  name                 = "unit6-subnet"
  resource_group_name  = azurerm_resource_group.rg.name
  virtual_network_name = azurerm_virtual_network.vnet.name
  address_prefixes     = ["10.0.1.0/24"]
}

resource "azurerm_public_ip" "public_ip" {
  name                = "unit6-public-ip"
  location            = azurerm_resource_group.rg.location
  resource_group_name = azurerm_resource_group.rg.name
  allocation_method   = "Static"
}

resource "azurerm_network_security_group" "nsg" {
  name                = "unit6-nsg"
  location            = azurerm_resource_group.rg.location
  resource_group_name = azurerm_resource_group.rg.name

  security_rule {
    name                       = "Allow-SSH"
    priority                   = 1001
    direction                  = "Inbound"
    access                     = "Allow"
    protocol                   = "Tcp"
```

```
  security_rule {
    name                       = "Allow-Web-8080"
    priority                   = 1002
    direction                  = "Inbound"
    access                     = "Allow"
    protocol                   = "Tcp"
    destination_port_range     = "8080"
    source_port_range          = "*"
    source_address_prefix      = "*"
    destination_address_prefix = "*"
  }
}

resource "azurerm_network_interface" "nic" {
  name                = "unit6-nic"
  location            = azurerm_resource_group.rg.location
  resource_group_name = azurerm_resource_group.rg.name

  ip_configuration {
    name                          = "internal"
    subnet_id                     = azurerm_subnet.subnet.id
    private_ip_address_allocation = "Dynamic"
    public_ip_address_id          = azurerm_public_ip.public_ip.id
  }
}

resource "azurerm_network_interface_security_group_association" "assoc" {
  network_interface_id      = azurerm_network_interface.nic.id
  network_security_group_id = azurerm_network_security_group.nsg.id
}
```

7. vm.tf

Defines the virtual machine.

Steps:

1. Create VM resource.
2. Specify image e.g. Ubuntu.
3. Attach NIC.
4. Add SSH key.

vm.tf - .../Desktop/Group_1

vm.tf

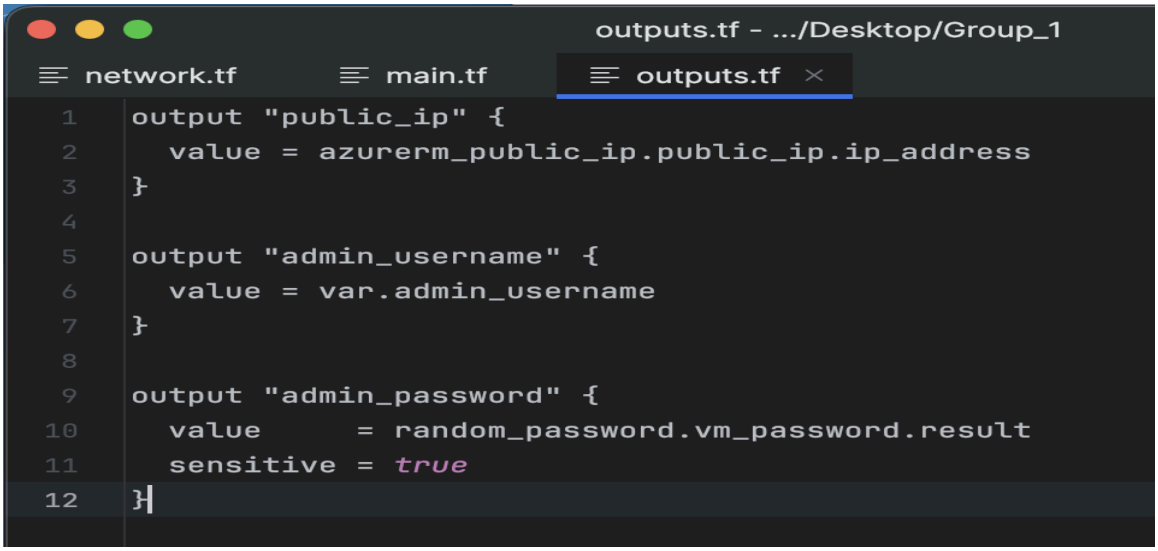
```
resource "random_password" "vm_password" {
  length  = 16
  special = true
}

resource "azurerm_linux_virtual_machine" "vm" {
  name                = "unit6-vm"
  resource_group_name = azurerm_resource_group.rg.name
  location            = azurerm_resource_group.rg.location
  size                = "Standard_D2s_v3"

  admin_username = var.admin_username
  admin_password = random_password.vm_password.result

  disable_password_authentication = false

  network_interface_ids = [
    azurerm_network_interface.nic.id
  ]

  os_disk {
    caching              = "ReadWrite"
    storage_account_type = "Standard_LRS"
  }

  source_image_reference {
    publisher = "Canonical"
    offer     = "ubuntu-24_04-lts"
    sku       = "server"
    version   = "latest"
  }
}
```

8. outputs.tf

Outputs important info.

Steps:

1. Output public IP.
2. Output VM name, user, password.



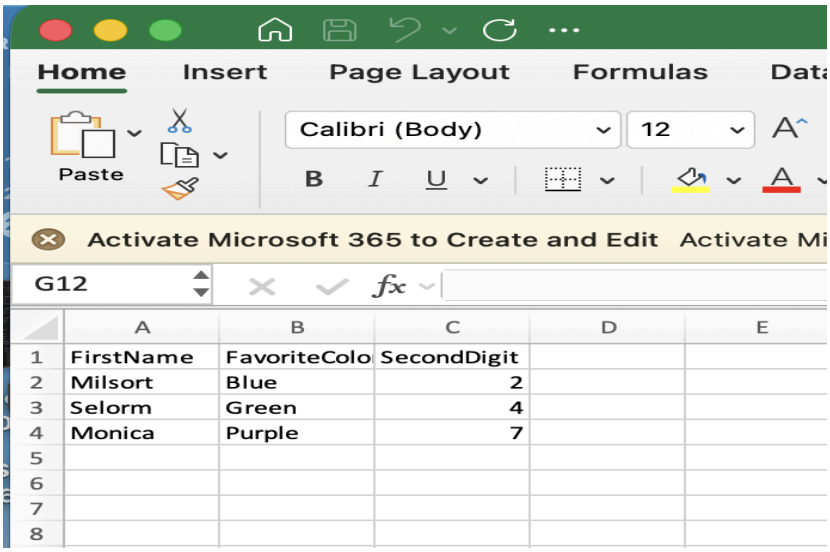

```
output "public_ip" {
  value = azurerm_public_ip.public_ip.ip_address
}

output "admin_username" {
  value = var.admin_username
}

output "admin_password" {
  value     = random_password.vm_password.result
  sensitive = true
}
```

9. students.csv

Stores student data.

Steps:

1. Add sample records.
2. Save in project folder.



10. Run Terraform

Deploy infrastructure.

Steps:

```
1. terraform init
2. terraform plan
3. terraform apply
```

```
(base) a1234@1234s-Laptop Group_1 % terraform state list
(base) a1234@1234s-Laptop Group_1 % cd /Users/a1234/Desktop
mkdir Group_1
cd Group_1
(base) a1234@1234s-Laptop Group_1 % touch main.tf variables.tf terraform.tfvars network.tf vm.t
f outputs.tf students.csv
(base) a1234@1234s-Laptop Group_1 % terraform init
terraform fmt
terraform validate
terraform plan
terraform apply
Initializing the backend...
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rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
Success! The configuration is valid.
```

11. SSH into VM

Connect to VM.

Steps:

1. Copy public IP.
2. Run ssh user@ip

```
(base) a1234@1234s-Laptop Group_1 % terraform output public_ip
"20.12.231.36"
(base) a1234@1234s-Laptop Group_1 % ssh Group_1@20.12.231.36
The authenticity of host '20.12.231.36 (20.12.231.36)' can't be established.
ED25519 key fingerprint is SHA256:F/Q0X3iKTlEtnKJdEokGTJIfo9/K/K6uaOGlCISCgKM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.12.231.36' (ED25519) to the list of known hosts.
Group_1@20.12.231.36's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1011-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Apr 27 16:59:21 UTC 2026

  System load:  0.07               Processes:               122
  Usage of /:   5.7% of 28.02GB    Users logged in:         0
  Memory usage: 3%                 IPv4 address for eth0: 10.0.1.4
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.
```

12. Start Web Server

Host CSV file.

Steps:

1. Copy students.csv to /var/www/html

```
Group_1@unit6-vm:~$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
136.32.253.13 - - [27/Apr/2026 17:20:26] "GET /students.csv HTTP/1.1" 200 -
136.32.253.13 - - [27/Apr/2026 17:21:23] "GET / HTTP/1.1" 200 -
136.32.253.13 - - [27/Apr/2026 17:21:24] code 404, message File not found
136.32.253.13 - - [27/Apr/2026 17:21:24] "GET /favicon.ico HTTP/1.1" 404 -
136.32.253.13 - - [27/Apr/2026 17:21:45] "GET /students.csv HTTP/1.1" 304 -
```

13. Browser Access

Access via web.

Steps:

1. Enter IP in browser.
2. View file.

```
Group_1@unit6-vm:~$ ls
students.csv
Group_1@unit6-vm:~$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

14. GitHub Setup

Version control (GitHub, 2024).

Steps:

1. Create repo.
2. git add .
3. git commit -m "Initial commit"
4. git push

```
(base) a1234@1234s-Laptop Group_1 % git add .
git commit -m "Initial commit"
git remote add origin git@github.com:KemboiM/Group-1.git
git branch -M main
git push -u origin main

[main f691af9] Initial commit
 Committer: 1234 <a1234@1234s-Laptop.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 .DS_Store
error: remote origin already exists.
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 446 bytes | 446.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:KemboiM/Group-1.git
   23edcef..f691af9  main -> main
branch 'main' set up to track 'origin/main'.
(base) a1234@1234s-Laptop Group_1 %
```

5. Team Contributions

Selorm

```
ser@DESKTOP-KV5B72U MINGW64 ~
 git clone https://github.com/KemboiM/Group-1.git
loning into 'Group-1'...
emote: Enumerating objects: 46, done.
emote: Counting objects: 100% (46/46), done.
emote: Compressing objects: 100% (33/33), done.
emote: Total 46 (delta 2), reused 16 (delta 1), pack-reused 0 (from 0)
eceiving objects: 100% (46/46), 13.04 KiB | 556.00 KiB/s, done.
esolving deltas: 100% (2/2), done.

ser@DESKTOP-KV5B72U MINGW64 ~
 cd C:/Users/user/Group-1

ser@DESKTOP-KV5B72U MINGW64 ~/Group-1 (main)
 nano students.csv

ser@DESKTOP-KV5B72U MINGW64 ~/Group-1 (main)
 git pull
emote: Enumerating objects: 11, done.
emote: Counting objects: 100% (11/11), done.
emote: Compressing objects: 100% (4/4), done.
emote: Total 8 (delta 5), reused 7 (delta 4), pack-reused 0 (from 0)
npacking objects: 100% (8/8), 817 bytes | 14.00 KiB/s, done.
rom https://github.com/KemboiM/Group-1
   b7a7a5b..23edcef  main       -> origin/main
pdating b7a7a5b..23edcef
ast-forward
 .gitignore | 1 +
 network.tf | 3 ++-
 2 files changed, 3 insertions(+), 1 deletion(-)

ser@DESKTOP-KV5B72U MINGW64 ~/Group-1 (main)
 git add vm.tf outputs.tf
```

Monica

```
AzureAD+SelormDarko@MID-PRJ-007 MINGW64 ~/Group-1 (main|MERGING)
$ git pull
git add main.tf variables.tf
git commit -m "Selorm configured Terraform provider and variables"
git push
error: You have not concluded your merge (MERGE_HEAD exists).
hint: Please, commit your changes before merging.
fatal: Exiting because of unfinished merge.
[main 23edcef] Selorm configured Terraform provider and variables
info: please complete authentication in your browser...
Enumerating objects: 9, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 581 bytes | 581.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/KemboiM/Group-1.git
   e4b28d6..23edcef  main -> main
```

Milsort

```
(base) a1234@1234s-Laptop Group_1 % git push -u origin main
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 10 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (13/13), 2.83 KiB | 2.83 MiB/s, done.
Total 13 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To github.com:KemboiM/Group-1.git
   df0389e..b7a7a5b  main -> main
branch 'main' set up to track 'origin/main'.
```

GIT

References

HashiCorp. (2023). *Terraform documentation*. https://developer.hashicorp.com/terraform/docs

Microsoft. (2024). *Azure virtual machines documentation*. https://learn.microsoft.com/azure/virtual-machines/

GitHub, Inc. (2024). *GitHub documentation*. https://docs.github.com/